

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Experiments(1 Question)

Course Code:	DSA0607	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 2– Experiments
Weightage:	40% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	<i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)</i>		
Student Name:	B Akshaya	Reg. No.:	192424336
Section / Batch:	2024/D	Date:	04.08.2026

Code of Conduct

I B Akshaya(192424336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B Akshaya

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

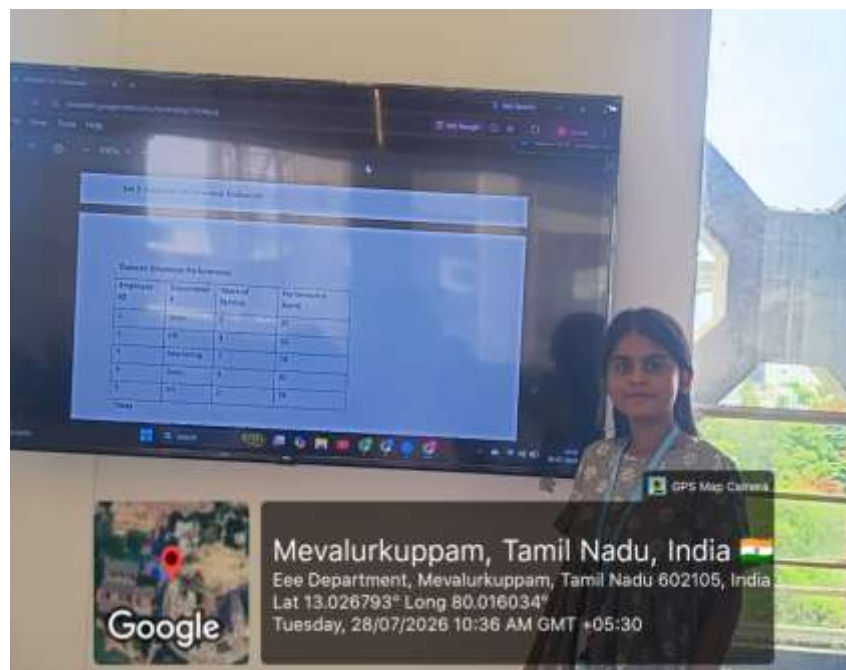
Section / Topic	Focus	Q Nos	Marks
Categorical Data Visualization	Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts	1	10
Distribution Analysis	Understanding defect distribution and variability using histograms, boxplots, and violin plots	2	10
Relationship Visualization	Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques	3	10
Geospatial & Temporal Visualization	Analyzing delivery performance using maps, time-series plots, and heatmaps	4	10
Distribution Comparison	Comparing patient data distributions using ridgeline plots, density plots, and boxplots	5	10
GRAND TOTAL:			50 Marks

Experiment 3: Time Series Trend and Seasonality Analysis

Objective: Analyze temporal patterns in sequential data.

Task:

- Collect time series data (e.g., monthly sales, temperature, website traffic).
- Plot line charts to observe trends over time.
- Decompose the series into trend, seasonal, and residual components.
- Identify patterns such as seasonality or irregular fluctuations.
- Compare raw data vs smoothed moving average.



Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context		
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts		
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution		
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools		
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____